

AI Excel Engine - Verification Report

Generated: 2026-01-03 15:57:58

Test Case #1: Ingest Excel File (Sanity Check)

Endpoint: /ingest/excel

Input Payload (Code/JSON):

```
{
  "file": "test_dataset_report.xlsx"
}
```

Engine Response:

```
{
  "status": "Ready",
  "file_path": "uploaded_files/test_dataset_report.xlsx",
  "columns": [
    "Staff ID",
    "Name",
    "Join Date",
    "Salary",
    "Dept"
  ]
}
```

Result: Captured

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Test Case #2: Advanced Analysis: Pivot Table

Endpoint: /analyze/

Input Payload (Code/JSON):

```
{
  "file_path": "uploaded_files/test_dataset_report.xlsx",
  "question": "Create a pivot table for 'Staff' sheet. Index='Dept', Values='Salary', aggfunc='mean'."
}
```

Engine Response:

```
{
  "type": "dataframe",
  "download_link": "uploaded_files/analysis_result.xlsx",
  "data": [
    {
      "Salary": 55000.0
    },
    {
      "Salary": 68500.0
    },
    {
      "Salary": 55000.0
    }
  ]
}
```

Result: Captured

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Test Case #3: Advanced Analysis: Date Calculation

Endpoint: /analyze/

Input Payload (Code/JSON):

```
{
  "file_path": "uploaded_files/test_dataset_report.xlsx",
  "question": "In 'Staff', add column 'Tenure' = difference in days between '2025-01-01' and 'Join Date'."
}
```

Engine Response:

```
{
  "type": "dataframe",
  "download_link": "uploaded_files/analysis_result.xlsx",
  "data": [
    {
      "Staff ID": 101,
      "Name": "Alice",
      "Join Date": "2023-01-01",
      "Salary": 50000,
      "Dept": "Sales",
      "Tenure": 731
    },
    {
      "Staff ID": 102,
      "Name": "Bob",
      "Join Date": "2023-06-15",
      "Salary": 60000,
      "Dept": "Sales",
      "Tenure": 566
    },
    {
      "Staff ID": 103,
      "Name": "Charlie",
      "Join Date": "2022-01-01",
      "Salary": 75000,
      "Dept": "IT",
      "Tenure": 1096
    },
    {
      "Staff ID": 104,
      "Name": "David",
      "Join Date": "2024-01-01",
      "Salary": 55000,
      "Dept": "HR",
      "Tenure": 366
    }
  ]
}
```

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```
    },  
    {  
      "Staff ID": 105,  
      "Name": "Eve",  
      "Join Date": "2023-03-10",  
      "Salary": 62000,  
      "Dept": "IT",  
      "Tenure": 663  
    }  
  ]  
}
```

Result: Captured

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Test Case #4: Advanced Analysis: Complex Filter

Endpoint: /analyze/

Input Payload (Code/JSON):

```
{
  "file_path": "uploaded_files/test_dataset_report.xlsx",
  "question": "Filter 'Staff' where Salary > 55000 and Dept is 'Sales'."
}
```

Engine Response:

```
{
  "type": "dataframe",
  "download_link": "uploaded_files/analysis_result.xlsx",
  "data": [
    {
      "Staff ID": 102,
      "Name": "Bob",
      "Join Date": "2023-06-15",
      "Salary": 60000,
      "Dept": "Sales"
    }
  ]
}
```

Result: Captured

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Test Case #5: V2 Feature: Sentiment Analysis Only

Endpoint: /enrich/sentiment

Input Payload (Code/JSON):

```
{
  "file_path": "uploaded_files/test_dataset_report.xlsx",
  "sheet_name": "Feedback",
  "text_column": "Comment",
  "tasks": [
    "sentiment"
  ]
}
```

Engine Response:

```
{
  "status": "Success",
  "rows_processed": 3,
  "download_link": "uploaded_files/dataset_updated.xlsx"
}
```

Result: Captured

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Test Case #6: V2 Feature: Summarization Only

Endpoint: /enrich/sentiment

Input Payload (Code/JSON):

```
{
  "file_path": "uploaded_files/test_dataset_report.xlsx",
  "sheet_name": "Feedback",
  "text_column": "Comment",
  "tasks": [
    "summary"
  ]
}
```

Engine Response:

```
{
  "status": "Success",
  "rows_processed": 3,
  "download_link": "uploaded_files/dataset_updated.xlsx"
}
```

Result: Captured

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Test Case #7: V2 Feature: Multi-Modal (Both)

Endpoint: /enrich/sentiment

Input Payload (Code/JSON):

```
{
  "file_path": "uploaded_files/test_dataset_report.xlsx",
  "sheet_name": "Feedback",
  "text_column": "Comment",
  "tasks": [
    "sentiment",
    "summary"
  ]
}
```

Engine Response:

```
{
  "status": "Success",
  "rows_processed": 3,
  "download_link": "uploaded_files/dataset_updated.xlsx"
}
```

Result: Captured

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Test Case #8: Advanced Analysis: Unpivot/Melt

Endpoint: /analyze/

Input Payload (Code/JSON):

```
{
  "file_path": "uploaded_files/test_dataset_report.xlsx",
  "question": "Melt the 'Staff' table. Keep 'Staff ID' and 'Name' as keys. Melt 'Salary' and 'Dept'."
}
```

Engine Response:

```
{
  "type": "dataframe",
  "download_link": "uploaded_files/analysis_result.xlsx",
  "data": [
    {
      "Staff ID": 101,
      "Name": "Alice",
      "Attribute": "Join Date",
      "Value": "2023-01-01"
    },
    {
      "Staff ID": 102,
      "Name": "Bob",
      "Attribute": "Join Date",
      "Value": "2023-06-15"
    },
    {
      "Staff ID": 103,
      "Name": "Charlie",
      "Attribute": "Join Date",
      "Value": "2022-01-01"
    },
    {
      "Staff ID": 104,
      "Name": "David",
      "Attribute": "Join Date",
      "Value": "2024-01-01"
    },
    {
      "Staff ID": 105,
      "Name": "Eve",
      "Attribute": "Join Date",
      "Value": "2023-03-10"
    }
  ]
}
```

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```
"Staff ID": 101,  
"Name": "Alice",  
"Attribute": "Salary",  
"Value": 50000  
,  
{  
  "Staff ID": 102,  
  "Name": "Bob",  
  "Attribute": "Salary",  
  "Value": 60000  
,  
{  
  "Staff ID": 103,  
  "Name": "Charlie",  
  "Attribute": "Salary",  
  "Value": 75000  
,  
{  
  "Staff ID": 104,  
  "Name": "David",  
  "Attribute": "Salary",  
  "Value": 55000  
,  
{  
  "Staff ID": 105,  
  "Name": "Eve",  
  "Attribute": "Salary",  
  "Value": 62000  
,  
{  
  "Staff ID": 101,  
  "Name": "Alice",  
  "Attribute": "Dept",  
  "Value": "Sales"  
,  
{  
  "Staff ID": 102,  
  "Name": "Bob",  
  "Attribute": "Dept",  
  "Value": "Sales"  
,  
{  
  "Staff ID": 103,  
  "Name": "Charlie",  
  "Attribute": "Dept",  
  "Value": "IT"  
,  
,
```

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```
{
  "Staff ID": 104,
  "Name": "David",
  "Attribute": "Dept",
  "Value": "HR"
},
{
  "Staff ID": 105,
  "Name": "Eve",
  "Attribute": "Dept",
  "Value": "IT"
}
]
```

Result: Captured

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Test Case #9: Edge Case: Invalid Column Name

Endpoint: /analyze/

Input Payload (Code/JSON):

```
{
  "file_path": "uploaded_files/test_dataset_report.xlsx",
  "question": "Show me the 'GhostColumn' in Staff."
}
```

Engine Response:

```
{
  "type": "dataframe",
  "download_link": "uploaded_files/analysis_result.xlsx",
  "data": [
    {
      "Staff ID": 101,
      "Name": "Alice",
      "Join Date": "2023-01-01",
      "Salary": 50000,
      "Dept": "Sales"
    },
    {
      "Staff ID": 102,
      "Name": "Bob",
      "Join Date": "2023-06-15",
      "Salary": 60000,
      "Dept": "Sales"
    },
    {
      "Staff ID": 103,
      "Name": "Charlie",
      "Join Date": "2022-01-01",
      "Salary": 75000,
      "Dept": "IT"
    },
    {
      "Staff ID": 104,
      "Name": "David",
      "Join Date": "2024-01-01",
      "Salary": 55000,
      "Dept": "HR"
    },
    {
      "Staff ID": 105,
      "Name": "Eve",

```

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```
"Join Date": "2023-03-10",  
"Salary": 62000,  
"Dept": "IT"  
}  
]  
}
```

Result: Captured

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Test Case #10: Edge Case: Missing File

Endpoint: /enrich/sentiment

Input Payload (Code/JSON):

```
{
  "file_path": "files/does_not_exist.xlsx",
  "sheet_name": "Sheet1",
  "text_column": "A"
}
```

Engine Response:

```
{
  "detail": "File not found"
}
```

Result: Captured